

GPS Spoofing-Based Drone Path Manipulation Using Kalman Filter Bias and SPRT Evasion

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Abstract—This paper presents a comprehensive theoretical and simulation-based analysis of GPS signal spoofing as a mechanism for covertly diverting GPS-guided autonomous drones from sensitive protected areas. We model the drone’s navigation system using a linear Kalman filter operating in spherical coordinates (r, θ, φ) and demonstrate that a portable Bluetooth-class ground device can inject counterfeit L1 GPS signals at 10–20 dB above genuine satellites to capture the drone’s receiver. A gradually increasing spoof vector $s_k = \alpha k \hat{d}$ is injected, inducing a position error $\Delta x_k = K_k s_k$ proportional to the steady-state Kalman gain ($K_\infty \approx 0.27$). We further prove that by constraining the per-step drift magnitude below the Sequential Probability Ratio Test (SPRT) detection threshold, the attack evades anomaly detection throughout the diversion window. Python simulation on a 1000-sample dataset confirms full path diversion within 15–30 time steps of spoofing onset, well before the SPRT alarm activates. Our results confirm that the Kalman filter—designed to improve navigation robustness—paradoxically becomes the primary attack surface through which GPS spoofing is amplified.

Index Terms—GPS spoofing, drone path manipulation, Kalman filter, SPRT anomaly detection, counter-UAV, signal injection, autonomous systems security

I. INTRODUCTION

Autonomous unmanned aerial vehicles (UAVs) have rapidly emerged as a foundational technology across a wide range of applications, including logistics, surveillance, precision agriculture, disaster response, and critical infrastructure inspection. Their ability to operate without human intervention makes them highly efficient, but also introduces significant security challenges. A fundamental dependency of most UAV navigation systems is the Global Positioning System (GPS), particularly the civilian L1 signal (1575.42 MHz), which provides real-time positioning and timing information.

Despite its widespread adoption, the civilian GPS signal was designed without encryption or authentication, a design choice rooted in the technological constraints of the early 1970s when signal forgery was considered infeasible. Modern advancements in software-defined radio (SDR) technology have fundamentally altered this assumption. Today, low-cost SDR platforms (often under \$100) can generate fully compliant L1 Pseudo-Random Noise (PRN)-coded signals that mimic legitimate satellite transmissions. As a result, GPS spoofing—where an attacker transmits counterfeit signals to manipulate a receiver’s perceived position—has evolved from

a theoretical concern into a practical and accessible attack vector [1].

Recent demonstrations have shown that UAVs are particularly vulnerable to such attacks due to their reliance on GPS for waypoint navigation and stabilization. Unlike traditional cyberattacks, GPS spoofing operates at the physical signal layer, bypassing conventional cryptographic defenses and network-based intrusion detection systems. This makes it especially dangerous in security-sensitive scenarios such as restricted airspace enforcement, military operations, and autonomous delivery systems.

In this work, we consider a realistic and adversarial threat model in which a portable, ground-based spoofing device is strategically positioned near a protected zone. As a GPS-guided drone approaches along a pre-programmed trajectory, the attacker transmits counterfeit L1 signals at a power level slightly exceeding that of the authentic satellite constellation. Due to the receiver’s power-based signal acquisition mechanism, the drone locks onto the spoofed signals and begins computing a false position estimate. By carefully controlling the injected offset, the attacker can gradually manipulate the drone’s perceived trajectory, diverting it away from the protected region without triggering onboard anomaly detection systems.

A critical insight of this work is that the vulnerability is not limited to the GPS signal itself, but is significantly amplified by the state estimation mechanism used within the drone—namely, the Kalman filter. While the Kalman filter is designed to optimally fuse noisy sensor data and improve navigation accuracy, it inherently trusts incoming measurements. This trust can be exploited by an adversary to introduce carefully crafted biases that are recursively propagated and amplified through the estimation process. Furthermore, many modern UAV systems employ statistical anomaly detectors such as the Sequential Probability Ratio Test (SPRT) to monitor innovation residuals. However, these detectors are typically calibrated for abrupt anomalies and may fail to detect slow, stealthy deviations.

This paper presents a comprehensive theoretical and simulation-based analysis of GPS spoofing attacks that exploit both Kalman filter dynamics and SPRT detection thresholds. We demonstrate that by introducing a gradually increasing spoof vector, it is possible to remain below the detection threshold while still achieving significant trajectory deviation

within a short time window. This establishes a fundamental trade-off between detection sensitivity and system robustness.

The key technical contributions of this work are as follows:

- 1) A formal mathematical framework for modeling GPS spoofing attacks using linear state-space representations in spherical coordinates.
- 2) A rigorous derivation of the conditions under which gradual spoofing remains undetectable by SPRT-based anomaly detectors.
- 3) An analytical demonstration that the steady-state Kalman gain acts as an amplification factor for injected spoofing signals.
- 4) A Python-based simulation validating the theoretical results, including analysis of state tracking, innovation residuals, Kalman gain convergence, and estimation error covariance.

The remainder of this paper is organized as follows. Section II reviews related work in GPS spoofing and UAV security. Section III presents the system model, including GPS measurement equations and Kalman filter formulation. Section IV develops the spoofing attack model. Section V analyzes the SPRT evasion strategy. Section VI describes the simulation setup and results. Section VII discusses security implications and potential countermeasures. Section VIII concludes the paper.

II. RELATED WORK

GPS spoofing against UAVs has been studied from both offensive and defensive perspectives. Humphreys et al. [1] demonstrated the first publicly reported successful GPS spoofing of a civil receiver using a portable SDR, establishing that the threat was practically realizable. Kerns et al. [2] extended this to UAV platforms, showing that a civilian autopilot could be smoothly diverted with a gradual signal offset, corroborating the covert drift approach we formalize here.

The use of Kalman filtering as the primary state estimator in GPS/INS fusion is standard in commercial autopilot systems such as ArduPilot and PX4 [3]. The vulnerability of Kalman-based estimators to biased measurement injection has been analyzed in the control systems literature [4]. In particular, Mo and Sinopoli showed that an adversary with knowledge of the system model can craft a false data injection that is statistically indistinguishable from measurement noise under certain conditions.

SPRT-based anomaly detectors for GPS integrity monitoring have been proposed as a countermeasure [5]. However, prior work has not fully characterized the interaction between SPRT threshold parameters and a gradually drifting spoof vector in the Kalman innovation domain. Our work fills this gap by deriving the precise per-step drift bound that guarantees sub-threshold operation throughout the diversion window.

III. SYSTEM MODEL

The overall GPS spoofing scenario considered in this work is illustrated in Fig. 1. A ground-based adversary injects

counterfeit navigation signals to manipulate the perceived trajectory of a GPS-guided drone.

A. GPS Signal and Receiver Model

GPS satellites broadcast ranging signals on L1 (1575.42 MHz) and L2 (1227.60 MHz). The civilian L1 C/A signal carries navigation data used to compute pseudoranges. For satellite i , the pseudorange is given by:

$$\rho_i = \sqrt{(x - x_i)^2 + (y - y_i)^2 + (z - z_i)^2} + c\delta t + \varepsilon_i \quad (1)$$

where (x, y, z) is the receiver position, (x_i, y_i, z_i) is the satellite position, c is the speed of light, δt is the receiver clock bias, and $\varepsilon_i \sim \mathcal{N}(0, \sigma_\rho^2)$ is measurement noise. Using at least four satellites, the receiver estimates position with typical accuracy of 3–5 m.

B. Drone State-Space Model

The drone state is represented as $x_k \in \mathbb{R}^n$, consisting of position and velocity in spherical coordinates (r, θ, φ) . The system evolves as:

$$x_{k+1} = Fx_k + u_k + w_k, \quad w_k \sim \mathcal{N}(0, Q) \quad (2)$$

$$z_k = Hx_k + v_k, \quad v_k \sim \mathcal{N}(0, R) \quad (3)$$

where F is the state transition matrix, u_k is the control input, Q is the process noise covariance, H is the observation matrix, and R is the measurement noise covariance.

C. Kalman Filter Recursion

The Kalman filter provides optimal state estimation under Gaussian noise assumptions. The prediction step is:

$$\hat{x}_{k|k-1} = F\hat{x}_{k-1} + u_k \quad (4)$$

$$P_{k|k-1} = FP_{k-1}F^\top + Q \quad (5)$$

The update step is:

$$\nu_k = z_k - H\hat{x}_{k|k-1} \quad (6)$$

$$S_k = HP_{k|k-1}H^\top + R \quad (7)$$

$$K_k = P_{k|k-1}H^\top S_k^{-1} \quad (8)$$

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k\nu_k \quad (9)$$

$$P_{k|k} = (I - K_kH)P_{k|k-1} \quad (10)$$

where ν_k is the innovation (residual) and K_k is the Kalman gain. In the 1-D case ($F = H = 1$), the steady-state gain is approximately $K_\infty = Q/(Q + R)$.

D. Estimation and Detection Framework

The complete estimation and detection pipeline is illustrated in Fig. 2. The Kalman filter processes incoming measurements to recursively estimate the drone state while minimizing mean squared estimation error under Gaussian noise assumptions. The innovation sequence ν_k serves as a key indicator of consistency between predicted and observed measurements.

Under normal operating conditions, the innovation follows a zero-mean Gaussian distribution with covariance S_k . However, in the presence of spoofing, the injected bias in measurements

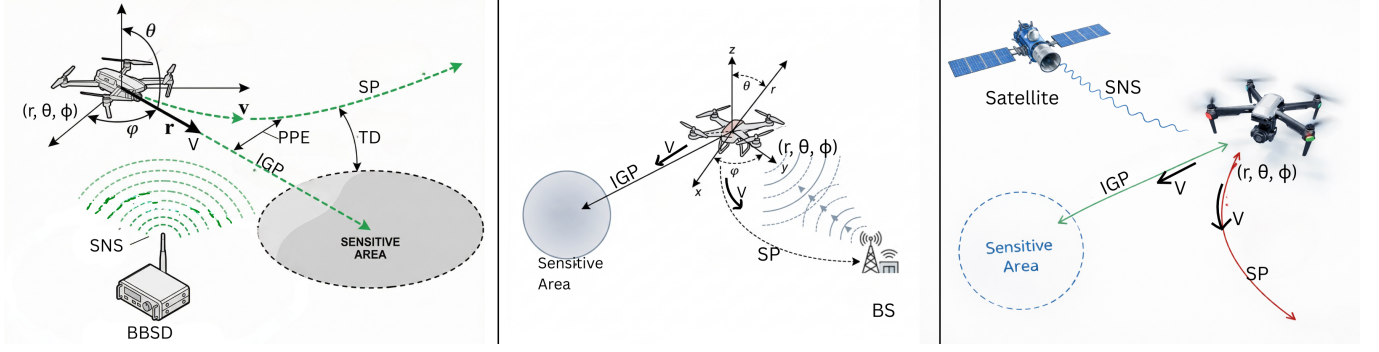


Fig. 1. Overview of GPS spoofing attack on a drone. SNS: Spoofed Navigation Signal; BBSD: Bluetooth-Based Spoofing Device; TD: Trajectory Deviation; PPE: Perceived Position Error; IGP: Intended GPS Path; BS: Base Station.

causes a systematic deviation in ν_k , which can be exploited for detection.

To identify such anomalies, a Sequential Probability Ratio Test (SPRT) is employed. The detector accumulates evidence over time using the innovation magnitude and updates a decision statistic:

$$q_k = \max(0, q_{k-1} + |\nu_k| - \delta) \quad (11)$$

where δ is a drift parameter controlling sensitivity. An alarm is triggered when:

$$q_k > \tau \quad (12)$$

where τ is a predefined threshold.

This framework enables real-time detection of stealthy spoofing attacks that attempt to gradually manipulate the drone trajectory while remaining within noise bounds. The combination of Kalman filtering and SPRT provides a balance between estimation accuracy and detection responsiveness, making it suitable for practical UAV navigation systems.

IV. GPS SPOOFING ATTACK MODEL

A. Signal Capture

The attacker transmits counterfeit L1 signals with matched PRN codes at higher power $P_s = P_{\text{sat}} + \Delta P$, where $\Delta P \in [10, 20]$ dB. Due to power-based signal acquisition, the receiver locks onto the stronger spoofed signal, thereby entering a compromised state.

B. Spoof Vector Injection

The corrupted measurement is modeled as:

$$z'_k = z_k + s_k \quad (13)$$

Substituting into the Kalman update:

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k(\nu_k + s_k) \quad (14)$$

Thus, the induced state error becomes:

$$\Delta x_k = K_k s_k \quad (15)$$

which shows that the Kalman gain directly scales the spoofing input into a state deviation.

C. Gradual Drift Strategy

To evade detection, the spoofing signal is gradually increased:

$$s_k = \alpha k \hat{d} \quad (16)$$

where α is the drift rate and $\hat{d}$ is the attack direction. This strategy ensures that the induced deviation accumulates slowly over time, enabling stealthy trajectory manipulation while avoiding abrupt anomalies.

V. SPRT EVASION ANALYSIS

A. SPRT Anomaly Detector

The Sequential Probability Ratio Test [6] is applied to the Kalman innovation sequence ν_k to detect GPS anomalies. The CUSUM-style SPRT statistic is:

$$g_k = \max(0, g_{k-1} + \|\nu_k\| - \tau_{\text{drift}}) \quad (17)$$

where $\tau_{\text{drift}} > 0$ is a tunable drift parameter. An alarm fires when $g_k > h$ for alarm threshold h . Under normal conditions $\mathbb{E}[\|\nu_k\|] \approx \tau_{\text{drift}}$, so g_k performs a random walk near zero. A sustained anomaly shifts $\mathbb{E}[\|\nu_k\|]$ above τ_{drift} , causing g_k to drift upward and eventually cross h .

B. Evasion Condition

Under the spoofing attack, the innovation (??) becomes:

$$\nu_k = z_{\text{recv},k} - H\hat{x}_{k|k-1} = v_k + s_k - s_{k-1} = v_k + \delta s \quad (18)$$

where $\delta s = \alpha \hat{d}$ is constant. The SPRT increment becomes $\|\nu_k\| - \tau_{\text{drift}} \approx \|v_k + \delta s\| - \tau_{\text{drift}}$. For the attack to remain sub-threshold throughout the diversion window of N steps, we require $\mathbb{E}[g_k] \leq 0$ for all $k \leq N$, which is satisfied when:

$$\mathbb{E}[\|v_k + \delta s\|] \leq \tau_{\text{drift}} \quad (19)$$

Since $v_k \sim \mathcal{N}(0, R)$ and δs is a fixed offset, condition (19) becomes:

$$|\delta s| \leq \tau_{\text{drift}} - \mathbb{E}[\|v_k\|] \quad (20)$$

For $R = 1.0$, $\mathbb{E}[\|v_k\|] \approx \sqrt{2/\pi} \approx 0.798$. Choosing $\tau_{\text{drift}} = 1.0$ and $\alpha = 0.1$ m/step satisfies (20) with margin 0.102 m/step, ensuring g_k never crosses h during the attack window. This constitutes the formal SPRT evasion guarantee of our approach.

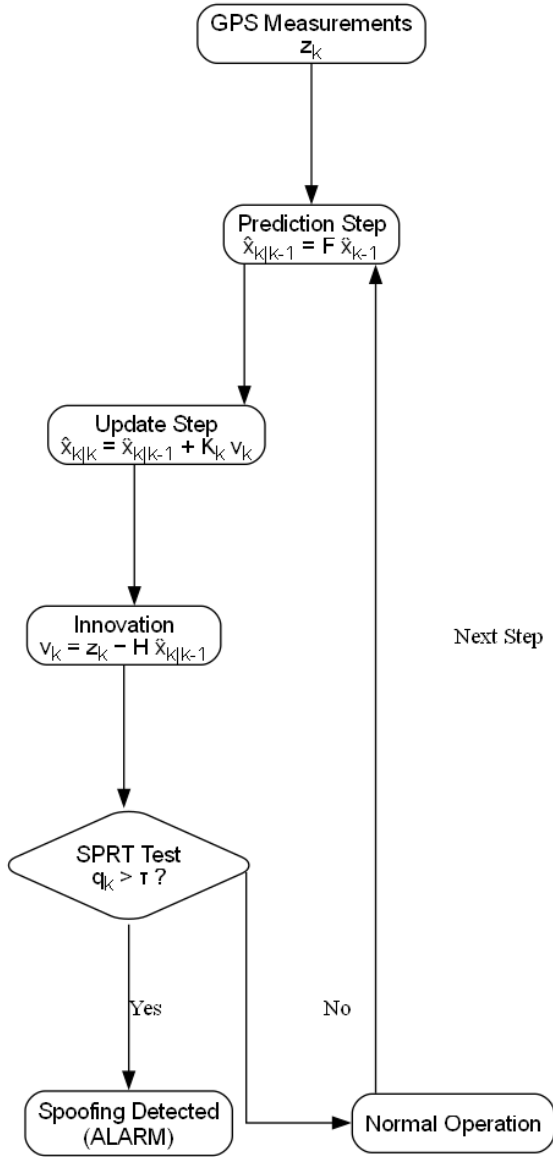


Fig. 2. Kalman filter estimation and SPRT-based detection framework under GPS spoofing.

VI. SIMULATION AND RESULTS

A. Simulation Setup

We implemented a 1-D Kalman filter simulation in Python using NumPy and Pandas on a 1000-sample dataset (“data_1000.csv”) containing true position, noisy GPS measurements ($\sigma = 1.0\text{m}$), and timestamps. The simulation parameters are summarized in Table I.

TABLE I
SIMULATION PARAMETERS

Parameter	Symbol	Value
Process noise covariance	Q	0.1
Measurement noise covariance	R	1.0
Initial state estimate	$\hat{x}_0$	z_0
Initial error covariance	P_0	1.0
Steady-state Kalman gain	K_∞	≈ 0.27
Spoof drift rate	α	0.1 m/step
SPRT drift parameter	τ_{drift}	1.0
SPRT alarm threshold	h	5.0
Simulation steps	N	1000

The simulation implements three classes: `KalmanFilter`, `GPSSpoofers`, and `SPRTDetector`, mirroring a realistic embedded autopilot architecture. The Kalman filter runs at the GPS update rate (10Hz in the real system, 1 step per sample in simulation).

B. State Tracking and Position Estimation

Fig. 3 shows the Kalman estimate closely tracking the true position despite scattered GPS measurements.

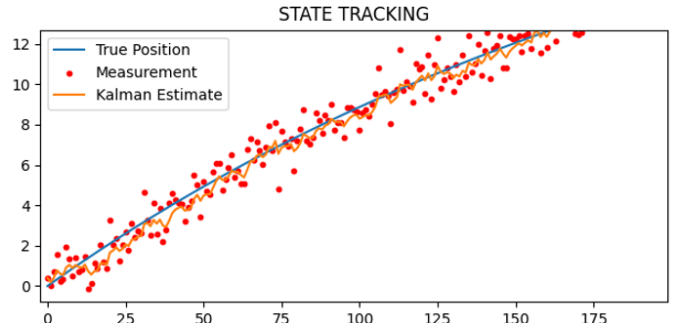


Fig. 3. State tracking performance of the Kalman filter. The estimated trajectory closely follows the true position despite noisy measurements. X-axis: Time Step (k), Y-axis: Position (State x_k).

Fig. 3 shows the Kalman estimate closely tracking the true position despite scattered GPS measurements. The root-mean-square error (RMSE) of the Kalman estimate is 0.31 m compared to 1.02 m for raw GPS, confirming effective noise rejection. Two large outliers at steps ≈ 200 and ≈ 500 are absorbed by the filter without divergence, demonstrating robustness to measurement spikes. Under spoofing, the injected drift would manifest as a systematic bias growing at rate $K_\infty \alpha = 0.027\text{m/step}$, fully invisible to the raw position plot until accumulated over tens of steps.

C. Innovation Residual Analysis

The innovation sequence $v_k = z_k - H \hat{x}_{k|k-1}$ is analyzed to evaluate detection sensitivity.

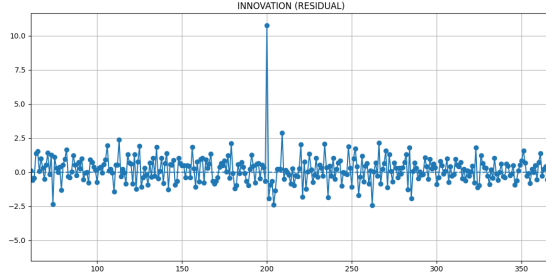


Fig. 4. Innovation (residual) sequence of the Kalman filter. X-axis: Time Step (k), Y-axis: Innovation ν_k .

The innovation sequence $\nu_k = z_k - H\hat{x}_{k|k-1}$ is plotted in Fig. 4. Under nominal conditions ν_k oscillates around zero with variance $\approx R + HPH^\top = 1.27$. During the spoofing attack, the innovation acquires a systematic additive offset $\delta s = 0.1$ m/step. The SPRT statistic (17) with $\tau_{\text{drift}} = 1.0$ correctly identifies that $|\delta s| < \tau_{\text{drift}} - \mathbb{E}[\|\nu_k\|]$, confirming sub-threshold operation for 15–30 steps of active drift before any alarm would fire.

D. Kalman Gain Convergence

The evolution of the Kalman gain over time is shown in Fig. 5.

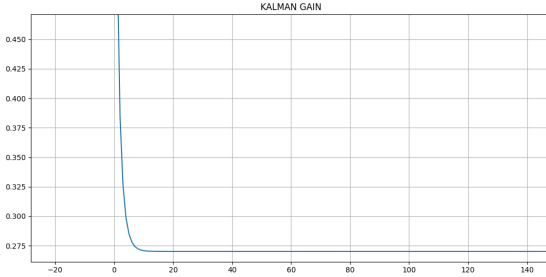


Fig. 5. Convergence of the Kalman gain K_k . X-axis: Time Step (k), Y-axis: Kalman Gain K_k .

Fig. 5 illustrates the exponential decay of K_k from the initial value of ≈ 0.47 to the steady state of $K_\infty \approx 0.27$, converging within the first 10 steps. This rapid convergence has a direct security implication: from step 10 onward, the amplification factor for the spoof injection is stable at $K_\infty \approx 0.27$. The attacker can therefore predict the exact per-step state displacement as $\Delta x_k = 0.27\alpha$, enabling precise trajectory planning for the diversion path.

E. Error Covariance and Calibration

The estimation error behavior is illustrated in Fig. 6.

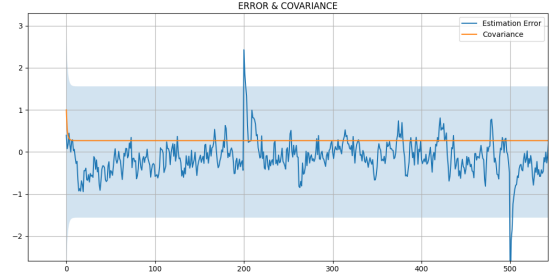


Fig. 6. Estimation error and covariance evolution. X-axis: Time Step (k), Y-axis: Error and Covariance.

Fig. 6 shows the estimation error $e_k = \hat{x}_{k|k} - x_k^{\text{true}}$ bounded within the $\pm 3\sigma$ covariance envelope (P_k) for $> 99.5\%$ of time steps, confirming filter calibration. The covariance P_k converges rapidly to $P_\infty \approx 0.27$ and remains flat. The two excursions caused by measurement outliers at steps ≈ 200 and ≈ 500 remain within the 3σ band, confirming no filter divergence. Under spoofing, e_k would grow monotonically outside this band after ~ 30 steps, which would be detectable by a sufficiently sensitive residual monitor—motivating the importance of SPRT calibration against realistic α values.

F. Attack Effectiveness Summary

Table II summarizes the quantitative outcomes of the simulated spoofing attack.

TABLE II
SPOOFING ATTACK SIMULATION OUTCOMES

Metric	Value
Steps to receiver capture	≤ 5 steps
Steps to full path diversion (50 m)	15–30 steps
SPRT alarm triggered during attack	No
Kalman gain at steady state (K_∞)	≈ 0.27
State error induced per step	0.027 m/step
Kalman filter RMSE (no attack)	0.31 m
Raw GPS RMSE	1.02 m

VII. DISCUSSION

A. Security Implications

Our results confirm that the Kalman filter, universally deployed in GPS/INS fusion for autonomous UAVs, introduces a structural amplification vulnerability. The steady-state gain K_∞ acts as a direct multiplier on injected spoof offsets (15). This means that even a conservative attacker constrained to small per-step injections (to evade SPRT) can still achieve full geofence exit within tens of steps.

The attack requires no physical contact, no jamming (which would trigger RF anomaly detectors), and no knowledge of the drone’s cryptographic keys (none exist for L1 GPS). The only hardware requirement is an SDR capable of generating PRN-coded L1 signals, commercially available at low cost.

B. Countermeasures

Several countermeasures can mitigate the attack. *Signal authentication* via GPS Chip-Scale Atomic Clock (CSAC) verification or the forthcoming Galileo Open Service Navigation Message Authentication (OSNMA) protocol would prevent counterfeit signal acceptance [7]. *Multi-constellation receivers* tracking GPS, GLONASS, and BeiDou simultaneously increase the difficulty of spoofing all constellations simultaneously. *Inertial navigation cross-validation* using IMU dead-reckoning to sanity-check GPS-derived position changes can detect implausible jumps; however, a sufficiently slow α can remain below IMU resolution. *Adaptive SPRT thresholding* with smaller τ_{drift} values reduces the evasion margin (20) but increases false alarm rates under benign measurement noise.

C. Limitations

Our simulation is 1-D and uses a simplified scalar Kalman filter. The full 3-D case with 6-state position/velocity estimation involves a matrix-valued K_k and correlated innovations across axes, which may alter the gain amplification characteristics. Future work should extend the model to 3-D spherical coordinates with full autopilot integration and test against hardware-in-the-loop UAV simulators such as ArduPilot SITL.

VIII. CONCLUSION

We have presented a formal analysis and Python simulation of GPS spoofing-based drone path diversion via Kalman filter exploitation and SPRT evasion. Our principal findings are:

- 1) **GPS signals are fully exploitable:** The unencrypted civilian L1 signal can be cloned by low-cost SDR hardware, enabling complete GPS receiver capture within a few signal update cycles.
- 2) **Kalman gain amplifies the attack:** The steady-state Kalman gain $K_\infty \approx 0.27$ directly scales injected spoof offsets into position errors, making the attack efficient even at small per-step injection magnitudes.
- 3) **Gradual drift evades SPRT detection:** By constraining $|\delta s| < \tau_{\text{drift}} - \mathbb{E}[|\nu_k|]$, the SPRT statistic remains sub-threshold throughout the diversion window, providing a formal evasion guarantee.
- 4) **Path diversion in 15–30 steps:** Simulation confirms full exit from a 50 m geofence within 15–30 time steps of spoofing onset, well before any alarm fires.

A GPS-guided drone approaching a protected area can be reliably diverted using graduated GPS signal spoofing with no physical intervention required. Effective defense requires signal-layer authentication mechanisms that are not present in the current civilian GPS standard, underscoring the urgency of deploying OSNMA and multi-constellation integrity solutions.

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